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MILITARY SPECIFICATION

ANCHORS, GUY AND RODS, ANCHOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE. This specification covers anchors and anchor rods for use in pole line and tower construction.

1.1 Anchor classification. Anchors shall be of the following types and classes:

Type I. Cone anchor.

Class A — 8-inch nominal hole diameter.

Class B — 10-inch nominal hole diameter.

Class C — 12-inch nominal hole diameter.

Type II. Expanding anchor.

Class A — 6-inch nominal, expanded area approx. 70 square inches.

Class B — 8-inch nominal, expanded area approx. 100 square inches.

Class C — 8-inch nominal, expanded area approx. 185 square inches.

Class D — 10-inch nominal, expanded area approx. 200 square inches.

Class E — 12-inch nominal, expanded area approx. 300 square inches.

Type III. Plate anchor.

	Approximate dimensions of plate in inches
Class A	6 × 22
Class B	8 × 22
Class C	8 × 35
Class D	10 × 40

Type IV. Screw anchor (no wrench).

Class A — 4-inch diameter.
Class B — 6-inch diameter.
Class C — 8-inch diameter.
Class D — 10-inch diameter.

Type V. Screw anchor (swamp).

Class A — 8-inch diameter.
Class B — 10-inch diameter.
Class C — 12-inch diameter.
Class D — 15-inch diameter.

FSC 4930

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Type VI. Rock anchors (expanding type, threaded rod).

Class A — ¾ × 15 inch.

Class B — ¾ × 30 inch.

Class C — 1 × 18 inch.

Type VII. Rock anchor (split bolt and wedge type with strand-eye).

Class A — 1-inch by 18 inches.

Type VIII. Rock anchor (straight rod threaded with strand-eye).

Class A — 1-inch by 12 inches.

Class B — 1-inch by 14 inches.

Type IX. Rock anchor (threaded rod with 45 degree strand-eye angular head).

Class A — 1-inch by 12 inches.

Class B — 1-inch by 14 inches.

1.2 Anchor rod classification. Anchor rods shall be of the following types and classes.

Type I. Threaded rods.

Class A — Strand-eye ½ inch diameter.

Class B — Oval-eye ¾ inch diameter.

Class C — Strand-eye ¾ inch diameter.

Class D — Three strand-eye ¾ inch diameter.

Class E — Oval-eye 1 inch diameter.

Class F — Strand-eye 1 inch diameter.

Class G — Three strand-eye 1 inch diameter.

Class H — Oval-eye 1¼ inch diameter.

Class I — Three strand-eye 1¼ inch diameter.

Type II. Plate anchor rods.

Class A — Oval-eye ¾ inch diameter.

Class B — Strand-eye ¾ inch diameter.

Class C — Three strand-eye ¾ inch diameter.

Class D — Oval-eye 1 inch diameter.

Class E — Strand-eye 1 inch diameter.

Class F — Three strand-eye 1 inch diameter.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL–P–3682 — Packaging of Hardware, Builders and Pole Line.

MIL–Z–17871 — Zinc-Coating (Hot-Dip Galvanizing).

STANDARDS

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–129 — Marking for Shipping and Storage.

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(Copies of specifications and standards required by the contractors in connection with specific procurement functions should be obtained from the procuring Agency or as directed by the contracting officers.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 **Preproduction samples.**

3.1.1 *Quantities.* After award of contract 6 anchors and 6 anchor rods of each type, class, and size, on order shall be furnished for preproduction inspection and test.

3.1.2 *Manufacture.* Preproduction samples shall be made and assembled by tools and methods that, as far as practicable, are the same as those which will be used for quantity production of the equipment.

3.1.3 *Compliance with specified requirements.* Preproduction samples shall be accompanied by a statement that they meet specified requirements for the equipment. Approval of preproduction samples shall not be construed as a waiver of any specified requirement.

3.1.4 *Disapproval.* When any samples are disapproved, the contractor will be required to submit additional samples until satisfactory samples have been submitted. Such modified samples shall be accompanied by a description of the changes which have been incorporated in the new samples in order to correct the faults of the rejected samples.

3.1.5 *Test data.* Preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations, and essential details of the testing equipment such as model and serial numbers and date of calibration.

3.1.6 *Reference standards.* After preproduction samples have been approved and returned to the contractors plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.2 **Materials and construction.** Materials and construction shall be as specified herein.

3.3 **Nut retainer.** Type I and type II anchors shall be provided with nut retainers. The nut retainer shall be constructed so as to prevent the nut from turning and riding up the anchor rod during installation of the anchor. It shall be possible to reclaim the anchor rod by removing it from anchor, if the anchor is to be abandoned.

3.4 **Design variations.** Minor variations in the design of anchors and anchor rods are permitted, provided, however, that no variation shall be permitted which would substantially modify their utility.

3.5 **Anchors when tested in accordance with paragraph 4, shall meet the requirements shown in table I.**

3.6 **Finish on anchors.** Unless hot-dip galvanized, all anchors shall be completely covered with a heavy coat of rust-resistant, asphalt paint.

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TABLE I. *Anchor Requirements*

Anchor type	Nominal size in in.	Anchor rod dia. in in.	Pull test in thousand lbs.	Rod length in in.	National pipe size
Type I, Class A ...	8 dia.	¾	16		
Class B ...	10 dia.	¾	16		
Class C ...	12 dia.	¾	16		
Type II, Class A ...	6 dia.	¾	16		
Class B ...	8 dia.	¾	16		
Class C ...	8 dia.	¾	16		
Class D ...	10 dia.	1	32		
Class E ...	12 dia.	1¼	48		
Type III, Class A ..	6 x 22	¾	19	96	
Class B ..	8 x 22	¾	19	96	
Class C ..	8 x 35	1	41	96	
Class D ..	10 x 40	1	41	96	
Type IV, Class A ..	4	¾	7	54	
Class B ..	6	¾	9	66	
Class C ..	8	1	13	66	
Class D ..	10	1¼	16	66	
Type V, Class A ..	8		7		1¼
Class B ..	10		11		1½
Class C ..	12		14		2
Class D ..	15		18		2
Type VI, Class A ..		¾	16	15	
Class B ..		¾	16	30	
Class C ..		1	40	18	
Type VII, Class A .		1	41	18	
Type VIII, Class A .		1	32	12	
Class B .		1	32	14	
Type IX, Class A ..		1	32	12	
Class B ..		1	32	14	

3.7 Anchor rod materials. Anchor rods shall be made of FS 1035 open-hearth, steel per Specification QQ-S-633. Anchor rods shall meet the ultimate load in 1000 pound requirements as shown in table II.

TABLE II. *Anchor Rod Requirements*

Rod style and shape	Eye shape	Rod dia. in in.	Rod length in feet	Ultimate load in 1,000 lbs.
Type I, Class A	Strand	½	8	13
Class B	Oval	¾	8	23
Class C	Strand	¾	8	23
Class D	3-Strand	¾	8	23
Class E	Oval	1	8	43
Class F	Strand	1	8	43
Class G	3-Strand	1	8	43
Class H	Oval	1¼	8	70
Class I	3-Strand	1¼	8	70
Type II, Class A	Oval	¾	8	32
Class B	Strand	¾	8	32
Class C	3-Strand	¾	8	32
Class D	Oval	1	8	56
Class E	Strand	1	8	56
Class F	3-Strand	1	8	56

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3.8 Screw threads. Threads on anchor rods shall conform to Handbook H–28 and shall be class 2 fit. Three-quarter inch diameter rods may have rolled threads. Larger diameter rods shall have cut threads. Rods shall be threaded a minimum of 3½ inches.

3.9 Finish on anchor rods. All anchor rods shall be hot-dip galvanized coated in accordance with Military Specification MIL–Z–17871.

3.10 Marking. The nominal sizes of anchors and anchor rods shall be clearly stamped in a readily accessible place. Characters shall be approximately three-sixteenths of an inch in height.

3.11 Workmanship. Anchors and anchor rods shall be manufactured and assembled in a thoroughly workmanlike manner. Parts shall not be bent, broken or otherwise impaired. Expanding anchors shall be operable in a normal manner without use of excess force.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection general.

4.1.1 *Definitions.* Standard MIL–STD–109 applies for definitions of inspection terms used herein.

4.1.2 *Classification of inspection.* Inspection shall be as follows:

(a) Preproduction inspection.

(b) Acceptance inspection.

4.2 Preproduction inspection.

4.2.1 *Preproduction inspection of anchors; guy and rods, anchor.* This inspection shall consist of the appropriate inspections specified in table III.

TABLE III. *Preproduction Inspection*

Inspection	Rqt. par.	Insp. par.
Visual and mechanical ...		4.4
Pull test	3.4	4.5
Tensile strength (anchor rods only)	3.7	4.6

4.3 Acceptance inspection.

4.3.1 *Acceptance inspection.* This inspection shall consist of Group A and Group C inspection as specified in 4.3.1.1 through 4.3.1.3.1.

4.3.1.1 *Group A inspection.* This inspection (including sampling) shall conform to table IV and Standard MIL–STD–105.

TABLE IV. *Group A Inspection*

Inspection	Insp. par.	AQL
Visual and mechanical ...	4.4	1%

4.3.1.2 *Group B inspection.* Not required.

4.3.1.3 *Group C inspection.* This inspection shall be as listed in table V.

4.3.1.3.1 *Sampling.* One anchor, guy or rod; anchor shall be selected from each production lot for Group C inspection.

4.3.1.3.2 *Noncompliance.* If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine cause of failure, the matter shall be referred to the contracting officer (see 6).

TABLE V. *Group C Inspection*

Inspection	Rqt. par.	Insp. par.
Pull test	3.5	4.5
Tensile strength (anchor rods only)	3.7	4.6

4.3.2 *Disposition of nonconforming product.* Disposition of lots and sample units rejected during group A inspection shall be in accordance with Standard MIL–STD–105.

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The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A or C inspection.

Inspection lots rejected during group A inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Visual and mechanical inspection. Anchors and anchor rods shall be examined for the following:

4.4.1 *Anchors.* Anchors shall be inspected to determine that they meet the dimensional requirements of table I and that they are coated completely with the asphaltum.

4.4.2 *Anchor rods.* Anchor rods shall be inspected to determine that they meet the dimensional requirements of table II. If rod is threaded, the threads shall be sharp and clean and the nut free-running. The coating shall be reasonably smooth, continuous, adherent, and substatntially free from injurious blisters, lumps, gritty areas, acid spots, dross warts, flux, and excess zinc on edges or other location that will interfere with the mechanical performance of the rod.

4.5 Pull test.

4.5.1 *Pull test inspection for anchors.* All anchors (except type VI, rock anchors) shall be inspected as follows:

4.5.1.1 Using a drum at least 25 inches high and of such diameter or dimensions as will allow a minimum margin of approximately 6 inches around the anchor to be tested. The bottom of the drum shall be of ¼ inch minimum thickness steel and sides of 0.156 inch minimum thickness steel. A hole shall be provided in the center of the bottom plate for the anchor rod. The hole shall be 1/16 inch greater in diameter than anchor rod. The drum shall be filled to ¾ of its depth with a mixture of 2 parts sawdust and 1 part wood shavings moistened to saturation. The anchor to be inspected shall be placed upside down in the drum (opened, if required) on the sawdust mixture and an anchor rod of the appropriate diameter forced up from the bottom of the drum through the sawdust mixture and secured to the anchor. The length of the anchor rod shall be governed by the tension testing machine used. The drum shall then be placed on top of a tension testing machine such as that made by Tindus Gleen and centered over its movable vise jaws. The anchor or pull rod shall be clamped in the vise jaws and a tension as specified in table I for the anchor under inspection shall be applied for a minimum of 1 minute. The anchor or pull rod used for this test shall be of adequate strength material to prevent a failure in the rod.

4.5.1.2 The anchor shall not be deformed beyond the extent that it may effect its holding power, or become fractured, and the anchor rod shall not become detached.

4.5.2 *Rock anchors* shall be inspected as follows:

4.5.2.1 *Expanding type.* The anchor shall be inserted in a hole in hard rock. The hole shall be between ⅛ and ¼ inch larger than the diameter of the closed anchor and 12 inches deep. The anchor shall be expanded by tightening the anchor rod by means of a bar through the eye. After tightening, the pull tension specified in table I shall be applied to the anchor and held for a minimum of one minute.

4.5.2.2 *Split bolt and wedge type.* The anchor shall be inserted in a hole in hard rock.

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The hole shall be between 1/8 and 1/4 inch larger than the diameter of the closed anchor and 12 inches deep. The anchor shall be expanded by driving the anchor rod as far into the hole as possible and apply a strain at an angle of 45° from the axis of the anchor rod. After tightening, the pull tension specified in table I shall be applied to the anchor and held for a minimum of one minute.

4.5.2.3 The anchor shall not slip or show signs of mechanical failure.

4.6 **Tensile strength (anchor rods only).** Anchor rods shall be inspected to determine that they meet the tensile strength requirements specified in table II for the particular type.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking shall be in accordance with Specification MIL-P-3682.

5.1.1 *Level A.* Anchors, guy, and anchor rods shall be preserved and packaged in accordance with Specification MIL-P-3682.

5.1.2 *Level C.* The complete equipment shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing.

5.2.1 *Level A.* Anchors and anchor rods shall be packed in accordance with the overseas shipment requirements of Specification MIL-P-3682.

5.2.2 *Level B.* Anchors and anchor rods shall be packed in accordance with the domestic shipment, service packing requirements of Specification MIL-P-3682.

5.2.3 *Level C.* Anchors and anchor rods shall be packed in accordance with the domestic shipment, commercial packing requirements of Specification MIL-P-3682.

5.3 **Marking.** Interior packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** The anchors and rods covered in this specification are intended for use in pole line construction and tower guying.

6.2 **Ordering data.** Invitations for bids and contracts should specify the following:

(a) Title, number and date of this specification.

(b) Type required.

(c) Government agency to which preproduction samples shall be submitted (3.1).

(d) Directions for marking and shipping preproduction samples (3.1).

(e) Place of final inspection.

(f) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States thereby incurs no responsibility nor any obligation whatsoever; and the fact that Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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